

Quantifying Uncertainty in Minimally Invasive Imaging Pipelines

Ulrich Abernathy, Quentin Whitlock, Valeria Aalto, Lucia Vasquez

Hollis Mountain Observatory

Section: Biomedical
Published: 2024-12-03
DOI: 10.5281/mjas.13.4.02
Keywords: cohort design; parameter sweep; ablation study

Abstract

This paper reports a systematic evaluation of immunomodulatory perfusion assays. The measurements reported here were collected over twelve independent trials under controlled conditions. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

1. Introduction

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Sensitivity to the smoothing window proved modest across the range we examined.

Neither of the alternative formulations improved on the simple additive model in our setting. The reduced model retains most of the explanatory power at a fraction of the computational cost. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses.

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. We release the processing scripts and intermediate artefacts to support independent verification.

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2. Related Work

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3. Materials and Methods

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4. Experimental Setup

Replicate agreement was high, suggesting the protocol is robust to small handling differences. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. We release the processing scripts and intermediate artefacts to support independent verification. The measurements reported here were collected over twelve independent trials under controlled conditions. Residual error grows approximately linearly with the number of coupled parameters. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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5. Results

The measurements reported here were collected over twelve independent trials under controlled conditions. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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6. Discussion

Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The measurements reported here were collected over twelve independent trials under controlled conditions.

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